Stitching procedure: two racks

1. Before you start

■ 1. Design: two racks. Final size 178.7 x 128.5 mm (7.03 x 5.06 in).

■ 2. Hoop: use the 200x200 hoop (200 x 200 mm), mounted as labelled. The design leaves 10 mm to the nearest hoop edge (minimum 10 mm).

■ 3. Totals: 55837 stitches, 6 thread colours, 5 thread changes. Estimated time about 103 min (93 min sewing at 600 spm plus about 2 min per thread change).

■ 4. Shopping and prep list:

- Thread 1: olive (#739a5f), suggested Madeira Polyneon 1968 Fair Green.
- Thread 2: dark green (#264725), suggested Robison-Anton Polyester 5849 Blue Spruce.
- Thread 3: royal blue (#3953a4), suggested Isacord Polyester 3332 Forget Me Not.
- Thread 4: white (#fcfcfb), suggested Isacord Polyester 0015 White.
- Thread 5: red (#e92024), suggested Isacord Polyester 1704 Candy Apple.
- Thread 6: black (#090b09), suggested Isacord Polyester 0020 Black.
- Bobbin thread: 60 wt pre-wound, white (light fabric; the back of the embroidery shows the bobbin colour).
- Needle: 75/11 embroidery needle (sharp point); 90/14 on heavy canvas or denim. Start every project with a fresh needle.
- Stabilizer: One layer of medium-weight tear-away under the fabric (two layers on light cotton or linen). Heavy canvas or denim is stable enough on its own with one layer.
- Temporary spray adhesive (505 or similar) if the garment cannot be hooped and must be floated.
- Small curved embroidery scissors for jump threads, a seam ripper, and a lint brush.

■ 5. File transfer: copy design.dst to a USB stick (FAT32) in a folder you will recognise, or send it with the Poolin Wi-Fi app (InStitch). Name it clearly, for example two_racks_179mm.dst, so it is easy to find on the touchscreen.

2. Hooping

■ 1. Decide where the design goes on the garment and mark its centre with a water-soluble pen or a pin. The design is centred on the hoop origin, so this mark is where the needle starts.

■ 2. Orientation: the design is 179 mm wide and 128 mm tall; the top of the design (negative y in the previews) is the top of the artwork. Mount the hoop as labelled and check on screen that the design is upright relative to the garment before stitching.

■ 3. Stabilizer: One layer of medium-weight tear-away under the fabric (two layers on light cotton or linen). Heavy canvas or denim is stable enough on its own with one layer. Hoop the fabric with the stabilizer underneath. Drum-tight, grain straight.

■ 4. Hoop the fabric with the stabilizer underneath: loosen the outer ring, lay the stabilizer then the fabric over the inner ring, press the outer ring down, then tighten. The fabric should be drum-tight but not stretched (pull gently on wovens only).

■ 5. Line up the hoop's centre marks with your centre mark. Attach the hoop to the machine.

■ 6. Load design.dst on the machine, check its position on screen, and run the trace/outline function so the foot travels around the design area: confirm it clears the hoop edge and lands where you want it. Adjust and trace again if needed.

■ 7. Set the speed no higher than about 600 spm for the first run; slow down for small details.

3. Thread-by-thread run sheet

One block per colour stop, in sewing order. The machine stops and trims at the end of each colour.

■ 1. Stop 1 of 6: load olive (#739a5f), Madeira Polyneon 1968 Fair Green. ■

- Thread the needle with this colour before pressing start. Hold the thread tail for the first few stitches.
- This colour stitches: 16 small shape(s) stitched as satin-style fills (2 to 3 mm across); 96 satin column(s), 1.2 to 2.4 mm wide, 1268 mm in total; bean (triple) running-stitch lines, 620 mm in total, for details thinner than 1.5 mm; 17 small patch(es) filling spots the main objects left uncovered.
- 10607 stitches, about 18 min.
- Watch: Narrow satin columns: watch for thread breaks or shredding; slow the machine if the thread frays.
- Watch: Small satin-style shapes: check the edges stay crisp and the shape is not pulling narrow.
- Watch: Outline lines: check they line up with the fills already stitched (registration). If they drift, the fabric slipped in the hoop.
- Watch: 136 jump(s) inside this colour: trim the floating threads by hand at the stop (centre 5 mm, centre 5 mm, upper left 29 mm, upper left 4 mm ...).
- At the stop: the machine trims the colour automatically; this colour also has 136 jump(s) inside it that the machine does NOT trim. Cut the floating threads on top close to the fabric before continuing.
- Thread change to stop 2 (dark green): cut the thread near the spool, pull the old thread out forward through the needle (never back up through the tension discs), put on the new spool, thread the path, use the automatic needle threader, and hold the tail for the first stitches.



■ 2. Stop 2 of 6: load dark green (#264725), Robison-Anton Polyester 5849 Blue Spruce. ■

- This colour stitches: 38 small shape(s) stitched as satin-style fills (2 to 4 mm across); 54 satin column(s), 0.9 to 3.5 mm wide, 663 mm in total; bean (triple) running-stitch lines, 1131 mm in total, for details thinner than 1.5 mm; 7 small patch(es) filling spots the main objects left uncovered.
- 7744 stitches, about 13 min.
- Watch: Narrow satin columns: watch for thread breaks or shredding; slow the machine if the thread frays.
- Watch: Small satin-style shapes: check the edges stay crisp and the shape is not pulling narrow.
- Watch: Outline lines: check they line up with the fills already stitched (registration). If they drift, the fabric slipped in the hoop.
- Watch: 151 jump(s) inside this colour: trim the floating threads by hand at the stop (lower 4 mm, lower 3 mm, lower 5 mm, lower 4 mm ...).
- At the stop: the machine trims the colour automatically; this colour also has 151 jump(s) inside it that the machine does NOT trim. Cut the floating threads on top close to the fabric before continuing.
- Check registration: the new colour should meet the previous ones without gaps; if it is shifted, re-tension the hoop before going on.
- Thread change to stop 3 (royal blue): cut the thread near the spool, pull the old thread out forward through the needle (never back up through the tension discs), put on the new spool, thread the path, use the automatic needle threader, and hold the tail for the first stitches.



■ 3. Stop 3 of 6: load royal blue (#3953a4), Isacord Polyester 3332 Forget Me Not. ■

- This colour stitches: 6 satin column(s), 1.9 to 1.9 mm wide, 634 mm in total.
- 3634 stitches, about 6.1 min.
- Watch: Narrow satin columns: watch for thread breaks or shredding; slow the machine if the thread frays.
- Watch: 5 jump(s) inside this colour: trim the floating threads by hand at the stop (centre 39 mm, upper 40 mm, upper left 95 mm, centre left 39 mm ...).
- At the stop: the machine trims the colour automatically; this colour also has 5 jump(s) inside it that the machine does NOT trim. Cut the floating threads on top close to the fabric before continuing.
- Check registration: the new colour should meet the previous ones without gaps; if it is shifted, re-tension the hoop before going on.
- Thread change to stop 4 (white): cut the thread near the spool, pull the old thread out forward through the needle (never back up through the tension discs), put on the new spool, thread the path, use the automatic needle threader, and hold the tail for the first stitches.



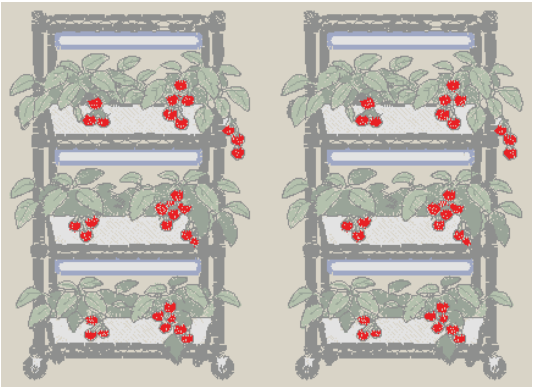
■ 4. Stop 4 of 6: load white (#fcfcfb), Isacord Polyester 0015 White.

• This colour stitches: 6 tatami fill area(s) totalling 1546 mm2 (largest: 324 mm2 upper left, 324 mm2 upper right, 279 mm2 centre right); 4 small shape(s) stitched as satin-style fills (2 to 3 mm across); 18 satin column(s), 1.2 to 3.9 mm wide, 466 mm in total; bean (triple) running-stitch lines, 88 mm in total, for details thinner than 1.5 mm; 8 small patch(es) filling spots the main objects left uncovered.
• 6531 stitches, about 11 min.
• Watch: Narrow satin columns: watch for thread breaks or shredding; slow the machine if the thread frays.
• Watch: Small satin-style shapes: check the edges stay crisp and the shape is not pulling narrow.
• Watch: Outline lines: check they line up with the fills already stitched (registration). If they drift, the fabric slipped in the hoop.
• Watch: 41 jump(s) inside this colour: trim the floating threads by hand at the stop (centre left 17 mm, centre left 5 mm, centre left 25 mm, centre left 10 mm ...).
• At the stop: the machine trims the colour automatically; this colour also has 41 jump(s) inside it that the machine does NOT trim. Cut the floating threads on top close to the fabric before continuing.
• Check registration: the new colour should meet the previous ones without gaps; if it is shifted, re-tension the hoop before going on.
• Thread change to stop 5 (red): cut the thread near the spool, pull the old thread out forward through the needle (never back up through the tension discs), put on the new spool, thread the path, use the automatic needle threader, and hold the tail for the first stitches.



■ 5. Stop 5 of 6: load red (#e92024), Isacord Polyester 1704 Candy Apple. ■

• This colour stitches: 62 small shape(s) stitched as satin-style fills (2 to 4 mm across).
• 2232 stitches, about 3.7 min.
• Watch: Small satin-style shapes: check the edges stay crisp and the shape is not pulling narrow.
• Watch: 49 jump(s) inside this colour: trim the floating threads by hand at the stop (lower right 4 mm, lower right 5 mm, lower right 17 mm, lower right 4 mm ...).
• At the stop: the machine trims the colour automatically; this colour also has 49 jump(s) inside it that the machine does NOT trim. Cut the floating threads on top close to the fabric before continuing.
• Check registration: the new colour should meet the previous ones without gaps; if it is shifted, re-tension the hoop before going on.
• Thread change to stop 6 (black): cut the thread near the spool, pull the old thread out forward through the needle (never back up through the tension discs), put on the new spool, thread the path, use the automatic needle threader, and hold the tail for the first stitches.



■ 6. Stop 6 of 6: load black (#090b09), Isacord Polyester 0020 Black. ■

- This colour stitches: 22 small shape(s) stitched as satin-style fills (2 to 3 mm across); 16 satin column(s), 1.7 to 2.9 mm wide, 1878 mm in total; bean (triple) running-stitch lines, 4203 mm in total, for details thinner than 1.5 mm; 20 small patch(es) filling spots the main objects left uncovered.
- 25089 stitches, about 42 min.
- Watch: Narrow satin columns: watch for thread breaks or shredding; slow the machine if the thread frays.
- Watch: Small satin-style shapes: check the edges stay crisp and the shape is not pulling narrow.
- Watch: Outline lines: check they line up with the fills already stitched (registration). If they drift, the fabric slipped in the hoop.
- Watch: 108 jump(s) inside this colour: trim the floating threads by hand at the stop (lower left 31 mm, lower left 19 mm, lower 18 mm, lower 6 mm ...).
- At the stop: the machine trims the colour automatically; this colour also has 108 jump(s) inside it that the machine does NOT trim. Cut the floating threads on top close to the fabric before continuing.
- Check registration: the new colour should meet the previous ones without gaps; if it is shifted, re-tension the hoop before going on.



4. Finishing

■ 1. Remove the hoop from the machine and take the fabric out of the hoop. Do not pull the fabric; loosen the ring first.

■ 2. Tear-away stabilizer: support the stitches with one hand and tear the stabilizer away gently, in small pieces, so the stitches are not distorted. Trim what remains close to the design.

■ 3. Trim remaining jump threads close on both the front and the back.

■ 4. Press from the back on a thick towel with a pressing cloth; never press the front directly (it flattens the thread).

5. Troubleshooting

Symptom	Likely cause	Fix	params.yaml setting
Puckering around fills	Not enough stabilizer, fabric hooped loose, or stitching too dense	Add a stabilizer layer, re-hoop drum-tight, reduce density	fill.row_spacing_mm (raise to 0.45), fill.underlay.*
Gaps between colours	Pull compensation too low for this fabric	Increase overlap and pull compensation, test again	general.overlap_mm (0.5), fill.pull_compensation_mm, satin.pull_compensation_mm
Outline does not line up with the fill	Fabric shifted in the hoop, or fills pulled in	Re-hoop with cut-away, more underlay, more compensation	fill.pull_compensation_mm, fill.underlay.edge_run_always

Symptom	Likely cause	Fix	params.yaml setting
Bird-nesting under the fabric	Top thread not seated in the tension discs or take-up lever; thread caught on the spool	Rethread the top from the start, check the spool unwinds freely	(not a digitizing problem)
Thread breaks or shredding	Dull or wrong needle, burr on the plate, too dense or too short stitches	New needle, slow the machine, open the density	satin.spacing_mm, fill.row_spacing_mm, general.min_stitch_length_mm
Bobbin thread showing on top	Top tension too tight or bobbin too loose	Lower top tension one step; check the bobbin case	satin.spacing_mm (if only on satins)
Satin text unreadable	Letters too small for the thread	Scale the design up or simplify the text	detail.min_text_height_mm, satin.min_width_mm, --size-mm
Underlay shows at the edges	Underlay inset too small for this fabric	Increase the underlay insets	fill.underlay.inset_mm, satin.underlay.contour_inset_mm
Design off-centre on the garment	Centre mark or hoop alignment off	Re-mark, re-hoop, use trace before stitching	(machine positioning)
Jump threads left on top	Single-needle machine does not trim inside a colour	Trim by hand at each stop	sequence.max_travel_mm, sequence.travel_under_cover

6. Test-stitch checklist

Stitch the design first on a scrap of the same fabric with the same stabilizer, then tick off:

■ 1. Edges are sharp and even

■ 2. No gaps between colours

■ 3. No underlay showing outside the top stitching

■ 4. No puckering around the fills

■ 5. Text and fine lines are readable

■ 6. Colours look as expected next to the garment

■ 7. Outline sits on the fill edges (registration)

■ 8. If anything fails, change the setting named in the troubleshooting table in params.yaml and re-export.